TC-Lab: what remains in the error?

Conductance variation, radiation and discrete measurements

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Work in progress. This layout draft contains audited intermediate results. Joint discrete-data fits and local information are audited. Profiles, refitted uncertainty calibration and retrospective transfer remain underway. It is not the completed study report.

Question	Evidence so far	Status
Can modest run variation help?	A 10% conductance band gives T1/T2 RMSE 0.605/0.118 °C in the joint bin-likelihood fit.	Audited; transfer and uncertainty pending.
Does radiation explain the rest?	Free radiation gives 0.590/0.119 °C in the joint fit; bin nll improves by 3.810.	Conditional change; boundary profile pending.
Are cold starts plausible?	Hidden heater offsets remain below 0.35 °C with either ±2 or ±5 °C bounds.	Working assumption, not verified equilibrium.
Are measurements discrete?	All 21,614 temperatures match a shifted 0.3223 °C code grid.	Hardware evidence; threshold phase remains assumed.
Do lag labels identify physics?	At the loading boundary, a nominal forced-response lag swap changes T1 by at most 0.00430 °C.	Weak assignment; lowering the loading bound barely changes the mean.
Are residuals adequate for inference?	Deterministic code diagnostics retain input, phase and cross-channel structure.	Whitening unfinished; no IID claim.

1 Records and model

We accept the same-device ENV2 provenance as the study assumption. The twelve distinct records comprise nine sine tests (three repeats at each of 30, 120 and 300 s) and three step tests. Duplicate aliases do not count as experiments. Only within-run time is available; Git history does not establish recovery time between runs.

Both temperatures enter a coupled heater/sensor model. With $i = 1, 2$ and the other heater denoted by k,

$$C_H \dot{H}_i = U_{a,j}(A_j - H_i) + U_s(S_i - H_i) + U_x(H_k - H_i) + \beta z_i, \tag{1}$$

$$C_S \dot{S}_i = U_s(H_i - S_i), \tag{2}$$

$$L \dot{z}_1 = u + c\,u(1 - u/100) - z_1, \qquad z_2 = 0. \tag{3}$$

The recorded command is held from its timestamp to the next update. The empirical curvature is not a measured electrical power law. Each run has $A_j = T_{1,j}(0) + a_j$, with $|a_j| \le 2\,^\circ\mathrm{C}$. The baseline starts $H_i(0) = S_i(0) = T_{i,j}(0)$ and $z_1(0) = 0$.

Absolute power calibration is unavailable. Estimate scale-free quantities

$$K_0 = \beta/U_{a,0}, \quad t_A = C_H/U_{a,0}, \quad r_S = C_S/U_s, \quad r = U_s/U_{a,0}, \quad q = U_x/U_{a,0}.$$

A constant shared heater calibration multiplier is absorbed into gain. Temperature-only fitting does not identify absolute conductances or heat capacities without an independent scale.

One physical coefficient varies across runs:

$$U_{a,j} = U_{a,0}e^{\delta_j}, \quad \sum_j \delta_j = 0, \quad \log(1-b) \le \delta_j \le \log(1+b).$$

Thus $K_j = K_0 e^{-\delta_j}$, $t_{A,j} = t_A e^{-\delta_j}$, $r_j = re^{-\delta_j}$ and $q_j = qe^{-\delta_j}$, while r_S stays shared. The hard band preserves coupling between experiments. It is not a population distribution or a fitted random-effects variance.

2 What the matched fits show

The old short-lag/curved-command coupled fit has T1 RMSE 1.022 °C. Orthogonal error projections attribute 74.6% of its T1 squared error to run offsets and another 19.3% to run trends. The additional fundamental sine projection contributes only 0.64%. These are descriptive projections, not independent physical causes. The earlier description of a one-degree amplitude error confused total RMSE with sine amplitude mismatch.

Gaussian comparison	T1 RMSE, °C	T2 RMSE, °C
Shared short lag and curved command	1.022	0.231
Conductance band 5%	0.646	0.147
Conductance band 10%	0.639	0.115
Conductance band 20%	0.641	0.115
10% band, affine ambient	0.728	0.111
10% band, hidden heater starts	0.615	0.111
10% convective band, free radiation	0.633	0.115

Widening the band adds little beyond 10% in this comparison. Two 10% deviations reach their lower bound; protocol-aligned ratios are fitted behavior, not measured airflow. Likelihood and unweighted RMSE need not rank models alike. These fits reuse all twelve records.

Radiation scale

Use Kelvin in $Q_{\mathrm{rad}} = \epsilon\sigma A_{\mathrm{rad}}(H^4 - T_r^4)$. For illustrative area 0.0012 m^2, surroundings at 23 °C and surface temperature 70.54 °C, emissivity scenarios 0.05–0.9 give 0.021–0.383 W. Against a linear radiative loss through ambient, matched at 50 °C, the nonlinear remainder is only 0.0021–0.0370 W. A locally linear radiative contribution can already reside in fitted U_a.

Metal alone does not determine emissivity. Finish, effective area, view factors and surface temperature are unknown; a sensor reading is not a known heater-surface temperature. Published TC-Lab models include convection, radiation and sensor dynamics [1]. Their values supply modeling context, not calibration for these records.

The free-radiation fit refits convection and gain to avoid double counting. Its normalized differential radiation coefficient at 50 °C is 0.0344 relative to nominal convection. Gaussian nll improves by 1.285. This is not a measured emissivity or a calibrated significance result.

Why assigning the two lags is difficult

At zero sensor loading, common/differential forced transfers factor as

$$G_{\pm}(s) = \frac{K}{(t_A s + 1 + \{0, 2q\})(1 + r_S s)(1 + Ls)}.$$

The output channels are $(G_+ + G_-)/2$ and $(G_+ - G_-)/2$. Exchanging r_S and L leaves these forced transfers unchanged in that limit.

At the fitted loading boundary $r = 0.005$, exchanging the nominal 14.98 s sensor lag and 2.69 s input lag changes the synthetic fast-sine T1/T2 waveforms by at most 0.00430/0.00185 °C. That calculation uses equal 20 °C starts and zero ambient offset. Unequal starts and ambient forcing can break the exchange. Repeated measurements can constrain sub-grid effects, so code spacing is not an absolute parameter-resolution limit. Lowering the positive loading bound from 0.005 to 0.0001 again reaches the bound and improves bin nll by only 0.01291; gain and fitted time scales barely change. This does not establish zero sensor heat capacity. Profiles must still test the lag assignments.

3 Discrete observations and residual checks

The hardware data, rather than just simulator output, are quantized. The inferred code values are $\mathrm{round}(0.3223n - 50, 2)$. Their grid does not identify pre-quantization noise or exact thresholds.

There is also a temperature-reference freedom in the linear model. When cold initial sensor states and ambient can shift together, changing nearest to floor bins translates every interval and predicted temperature by half a code, 0.16115 °C, without changing the likelihood. An independent synthetic check verifies this identity with slack ambient bounds. Fixed initial states, active ambient limits or absolute-temperature radiation can break it. A threshold phase comparison must distinguish this reference shift from dynamic information.

For the unreported analogue temperature,

$$Y_i^* = m_i(\psi) + X_i + v_i, \qquad X_i = e^{-\Delta t_i/\ell} X_{i-1} + w_i,$$

with stationary variance S, process variance $S(1 - e^{-2\Delta t_i/\ell})$ and independent $v_i \sim N(0, R)$. The likelihood integrates conditional observation density over each code bin. Forward quadrature and backward smoothing account for correlated latent errors. Multiplying marginal bin probabilities gives a different likelihood.

Holding the 10% mean fit fixed, bin-likelihood refitting changes analogue measurement SD from about 0.094 to 0.0255 °C for T1, and from 0.072 to 0.0325 °C for T2. T2 correlation length changes from 7.9 to 54.8 s. These are conditional noise estimates, not calibration measurements. The joint 10% bin-likelihood fit is now independently audited. Its T1/T2 RMSE is 0.605/0.118 °C, with $K_0 = 0.8732$ °C/% and $t_A = 174.59$ s. Gain is within 0.1% of the matched Gaussian estimate. The fitted noise is

Joint bin fit	T1	T2
Latent correlated SD, °C	0.5362	0.0681
Correlation length, s	658.28	62.56
Analogue measurement SD, °C	0.0258	0.0325

The joint 5% and 20% bands are also audited, with RMSE 0.637/0.153 °C and 0.606/0.118 °C. Their nll values are 6279.4250 and 6238.2167, versus 6238.2987 for 10%. Four 5% deviations reach bounds; the wider band is not selected from training likelihood alone. The joint radiation fit gives 0.590/0.119 °C and nll 6234.4890. Its improvement is larger than in the Gaussian approximation, but is neither a calibrated significance result nor evidence of measured emissivity.

Fisher's score identity supplies derivatives from smoothed latent moments [2]. Selected points require independent physical propagation, Gaussian bin rectangles, grid/support convergence and score stationarity. A repaired rare-bin integration failure remains a numerical failure; it is not evidence against the shared physical model.

Whitening checks combine independent randomized predictive residuals [3], serial and cross-channel dependence, input/phase dependence, and nonrandomized down/same/up code probabilities. Fifty fixed-model simulations are complete with no failed draw. Across twenty independent randomizations, median maximum serial correlation over lags 1–60 is 0.0248/0.0245 for T1/T2; maximum cross-channel correlation has median 0.0224. These are comparable to the fixed-model references. T1's nonrandomized code-transition discrepancy has reference upper-tail rank 0.0588. However, deterministic conditional expected Gaussian-PIT scores expose structure that the added PIT jitter diluted:

Statistic	Data	Reference 97.5%
T1 input/score correlation magnitude	0.04429	0.02261
T1 maximum phase projection	0.02983	0.01971
T2 maximum ACF, lags 1–60	0.04695	0.04039
Cross-channel maximum, lags ±10	0.04895	0.02540

Each descriptive upper-tail rank is 1/51. These scores are discrete and not unit-variance Gaussian residuals. The same calculation is applied to all fifty references. Removing run trends and three sine harmonics leaves cross correlation 0.04719, above its matched reference 97.5% value 0.02603. Projection has not resolved the dependence or identified its cause. Fixed-model references do not refit parameters or repeat selection, so these ranks are not calibrated composite-null p-values. Whitening remains unfinished.

Continuous-Gaussian joint sensitivities give process-driving correlation 0.0908 or measurement correlation −0.00692 when estimated separately. Estimating both gives 0.11587/−0.02463 and improves Gaussian nll by 6.14891. All have independent dense-covariance audits. This is a comparison on discrete temperatures using a continuous likelihood; it does not establish calibrated joint-bin probabilities or an environmental mechanism.

4 Uncertainty, transfer and open questions

The independently audited information matrix distinguishes the centered tangent space from the selected active constraint face. Conditional local standard errors are

Coordinate	IID analogue	Correlated analogue	Joint bin
$\log K$	0.0007263	0.0033010	0.0033620
$\log t_A$	0.0005152	0.0030056	0.0030888
$\log L$	0.0479381	0.0309354	0.0324303

The first two columns hold covariance known and use identical analogue marginal variance. Correlation increases gain and thermal-time SEs by 4.55 and 5.83, while reducing the short-lag SE to 0.645 of its IID value. The bin column also estimates noise nuisances; its difference does not isolate quantization. These are selected-basin, active-face approximations, not calibrated intervals.

T1 log latent SD and log correlation length have local correlation 0.9845. Conditioning the residual process on the initial code changes their estimates from 0.5362 °C/658.3 s to 0.8763 °C/1761.9 s, while $2S/\ell$ changes only 0.22% and the mean parameters barely change. The data constrain the diffusion rate more clearly than its stationary variance/length split.

Allowing each cold heater/sensor pair within its first reported bin adds 24 bounded coordinates. Its audited nll falls to 6209.5536, gain to 0.8697 °C/% and short lag to 2.72054 s; several initial states reach edges. T1 free-simulation RMSE increases to 0.655 °C, while T2 falls to 0.113 °C. Better correlated-bin likelihood does not

imply better noise-free simulation. These bin bounds do not integrate an initial measurement-noise posterior.

Profiles refit all ambient, conductance and noise nuisances. Sensor-loading bounds, initial-state sensitivities and lag alternatives make routine Wald intervals unsafe without those checks.

The separate fifty-draw recovery study refits simulated quantized records. Each draw also fits gain at its generating value to assess coverage of the nominal scalar likelihood-ratio cutoff. Failed fits, retries and Monte Carlo uncertainty remain in the denominator and report. This conditional exercise does not protect against mean/noise misspecification.

Retrospective leave-run/protocol-out fits will predict omitted records without fitting their ambient or conductance from later test temperatures. Show nominal simulation, hard-nuisance scenarios, noise ranges and parameter sensitivity separately. All records informed model selection, so these are internal transfer checks. Marginal coverage fractions do not turn correlated readings into independent validation replicates.

The remaining questions are whether the discrete-data model removes input-linked residual structure, whether the conductance band transfers across protocols, how startup/threshold assumptions assign the lag labels, and which uncertainty statements survive refitting and simulation. A future hood experiment should log ambient temperature, absolute time, prior run recovery, airflow settings, firmware and electrical power. A hood can add forced or fluctuating convection; it does not automatically supply a constant environment. No new hardware or DoE is part of this study.

Evidence and status

This draft uses terminal, independently audited intermediate artifacts in the committed remaining-error study. The full decision-gate audit, final figures and parameter-inclusive uncertainty/transfer results are pending. It will be replaced after those gates are assessed.

References

[1] Park, Martin, Kelly and Hedengren (2020). Benchmark temperature microcontroller for process dynamics and control. *Computers & Chemical Engineering*, article 106736. https://doi.org/10.1016/j.compchemeng.2020.106736.

[2] Poyiadjis, Doucet and Singh (2011). Particle approximations of the score and observed information matrix in state space models with application to parameter estimation. *Biometrika*, 98, 65–80. https://www.stats.ox.ac.uk/~doucet/poyiadjis_doucet_singh_particlescoreparameterestimation.pdf.

[3] Dunn and Smyth (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics*, 5, 236–244. https://gksmyth.github.io/pubs/residual.pdf.